

FINITE CONGRUENCE COVERINGS FOR THE ERDŐS–STRAUS EQUATION AND STERILE PROFINITE POINTS

ANONYMOUS

ABSTRACT. ABSTRACT-PLACEHOLDER

1. INTRODUCTION

INTRO-PLACEHOLDER

2. COVERINGS AS CLOPEN CLASSES

2.1. The seven families. We use the families of Elsholtz and Tao [2, Prop. 1.9]. Each is indexed by a triple κ of positive integers subject to a side condition, and defines a set of residue classes:

family	parameters	side condition	class
I1	a, d, f	$f \mid 4a^2d + 1$	$n \equiv -f \pmod{4ad}$
I2	a, c, f	$(4ac, f) = 1$	$n \equiv -f \pmod{4ac}, n \equiv -c/a \pmod{f}$
I3	c, d, f	$(4cd, f) = 1$	$n \equiv -f \pmod{4cd}, n^2 \equiv -4c^2d \pmod{f}$
I4	a, b, e	$e \mid a + b, (e, 4ab) = 1$	$n \equiv -1/e \pmod{4ab}$
II1	a, b, e	$e \mid a + b, (e, 4ab) = 1$	$n \equiv -e \pmod{4ab}$
II2	a, d, f	$4ad \mid f + 1$	$n \equiv -4a^2d \pmod{f}$
II3	a, d, e	$(4ad, e) = 1$	$n \equiv -4a^2d - e \pmod{4ade}$

On its class each family carries explicit solutions. With $\pi^I(a, b, c, d) = (abdn, acd, bcd)$ and $\pi^{II}(a, b, c, d) = (abd, acdn, bcdn)$ [2, §2], the remaining coordinates are

I1	$e = (4a^2d + 1)/f, b = (ne + 1)/(4ad), c = (n + f)/(4ad)$
I2	$d = (n + f)/(4ac), b = (na + c)/f$
I3	$a = (n + f)/(4cd), b = (n + (n^2 + 4c^2d)/f)/(4cd)$
I4	$c = (a + b)/e, d = (ne + 1)/(4ab)$
II1	$c = (a + b)/e, d = (n + e)/(4ab)$
II2	$c = (f + 1)/(4ad), b = (nc + a)/f$
II3	$c = (n + 4a^2d + e)/(4ade), b = ce - a = (n + e)/(4ad)$

and the solution is π^I for I1–I4 and π^{II} for II1–II3.

Lemma 2.1 (PROVED; [2, §§2, 10]). *Let κ be an admissible parameter triple of one of the seven families and let $n \geq 1$ be an integer in its class. Then the*

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coordinates above are positive integers, and the resulting (x, y, z) is a solution of $4/n = 1/x + 1/y + 1/z$ in positive integers.

Proof. The identity $4/n = 1/x + 1/y + 1/z$ follows from the defining relations of the varieties $\Sigma_n^I, \Sigma_n^{II}$ of [2, (2.1)–(2.9), (2.13)–(2.21)]. Integrality on the class is a direct check; for example in I3, $n \equiv -f \pmod{4cd}$ gives $a \in \mathbb{Z}$, the second congruence gives $Y := (n^2 + 4c^2d)/f \in \mathbb{Z}$, and $fY \equiv f^2 \pmod{4cd}$ with f invertible modulo $4cd$ gives $Y \equiv f \equiv -n$, so $b \in \mathbb{Z}$. With κ fixed, every coordinate is a polynomial in n with non-negative coefficients that is positive at $n = 1$, so all coordinates are positive for $n \geq 1$. $\square$

The positivity statement for all $n \geq 1$ (not only for large n) was checked in the project's review of [12]; the degrees of (x, y, z) in n are at most 4.

ET call a primitive residue class *solvable by polynomials* if there are polynomials P_1, P_2, P_3 with rational coefficients, taking positive integer values at all sufficiently large n of the class, with $4/n = \sum 1/P_i(n)$ identically. Lemma 2.1 says that every primitive class contained in one of the seven families is solvable by polynomials. We use the converse in the following form.

Cited theorem 2.2 ([2, Prop. 1.9]). If a primitive residue class is solvable by polynomials, then all sufficiently large primes in it lie in the union of finitely many classes of the seven families.

(ET state this separately for Type I and Type II solvable classes, with the families I1–I4 and II1–II3 respectively; every solvable primitive class is of one of the two types [2, p. 8].)

2.2. Profinite language. Let $\widehat{\mathbb{Z}} = \prod_q \mathbb{Z}_q$ be the profinite completion of $\mathbb{Z}$, with unit group $\widehat{\mathbb{Z}}^\times = \prod_q \mathbb{Z}_q^\times$. Every class of the table is defined by congruences modulo an integer $M(\kappa)$ (for I3, modulo $4cdf$); we write $\text{Cl}(\kappa) \subset \widehat{\mathbb{Z}}$ for the set of $x \in \widehat{\mathbb{Z}}$ satisfying them. It is open and closed (*clopen*), and an integer n lies in $\text{Cl}(\kappa)$ exactly when it lies in the class. We call $\text{Cl}(\kappa)$ an *ET class*.

Definition 2.3. A point $x \in \widehat{\mathbb{Z}}^\times$ is *sterile* if it lies in no ET class. For a subfamily $\mathcal{K}$ of parameter triples, x is $\mathcal{K}$ -*sterile* if it lies in no $\text{Cl}(\kappa)$, $\kappa \in \mathcal{K}$.

Let $\mathcal{P}$ be an infinite set of primes and $\mathcal{P}'$ its set of accumulation points in $\widehat{\mathbb{Z}}$.

Lemma 2.4 (PROVED). $\mathcal{P}'$ is a compact subset of $\widehat{\mathbb{Z}}^\times$. If $\Sigma \subset \widehat{\mathbb{Z}}^\times$ is clopen, defined modulo M (a union of reduced classes modulo M), and $\mathcal{P}(\Sigma)$ is the set of primes $p \nmid M$ with $p \bmod M$ in $\Sigma \bmod M$, then $\mathcal{P}(\Sigma)' = \Sigma$.

Proof. If $x \in \mathcal{P}'$, then for each prime q and each k infinitely many $p \in \mathcal{P}$ satisfy $p \equiv x \pmod{q^k}$; all but one of them are prime to q , so $x_q \in \mathbb{Z}_q^\times$. Derived sets in the compact metric space $\widehat{\mathbb{Z}}$ are closed, hence compact. For the second statement, a point of $\mathcal{P}(\Sigma)'$ is congruent modulo M to some

element of $\mathcal{P}(\Sigma)$, so it lies in Σ . Conversely every neighbourhood $x + Q\widehat{\mathbb{Z}}$ ($M \mid Q$) of $x \in \Sigma$ is a reduced class modulo Q and contains infinitely many primes by Dirichlet’s theorem; they lie in $\mathcal{P}(\Sigma)$. $\square$

The sets of primes studied below are all of the form $\mathcal{P}(\Sigma)$. The prime p itself is never a unit of $\mathbb{Z}_p$, which is why we work with accumulation points rather than with the primes as points of $\widehat{\mathbb{Z}}$.

Proposition 2.5 (PROVED). *Let $\mathcal{P}$ be an infinite set of primes.*

- (a) *If no point of $\mathcal{P}'$ is sterile, then finitely many ET classes contain all but finitely many $p \in \mathcal{P}$. For those p , Lemma 2.1 gives an explicit solution of $4/p = 1/x + 1/y + 1/z$.*
- (b) *If some $x \in \mathcal{P}'$ is sterile, then no finite set of ET classes contains all but finitely many $p \in \mathcal{P}$. Moreover no finite set of primitive residue classes, each solvable by polynomials, contains all but finitely many $p \in \mathcal{P}$.*

Hence all but finitely many elements of $\mathcal{P}$ lie in finitely many primitive classes solvable by polynomials if and only if $\mathcal{P}'$ contains no sterile point.

Proof. (a) The ET classes are open and cover the compact set $\mathcal{P}'$, so finitely many of them cover it; let U be their union. If $\mathcal{P} \setminus U$ were infinite, it would have an accumulation point, which lies in $\mathcal{P}' \subset U$; but U is open and misses $\mathcal{P} \setminus U$, a contradiction.

(b) Removing finitely many elements from $\mathcal{P}$ does not change $\mathcal{P}'$. If finitely many ET classes contain a cofinite subset of $\mathcal{P}$, their union is closed and therefore contains x , which contradicts sterility. Next let $C_1, \dots, C_s$ be primitive classes solvable by polynomials whose union contains a cofinite subset of $\mathcal{P}$. Viewed in $\widehat{\mathbb{Z}}$, their union is closed, so $x \in C_j$ for some j . By Cited Theorem 2.2 there are ET classes $K_1, \dots, K_t$ containing all large primes of C_j . Their union is closed and misses x , so some neighbourhood $V = x + Q\widehat{\mathbb{Z}} \subset C_j$ misses it. As x is a unit, V is a reduced class modulo Q , and by Dirichlet’s theorem it contains infinitely many primes. These lie in C_j and in no K_i , a contradiction. The final equivalence combines (a), (b) and Lemma 2.1. $\square$

Remark 2.6 (Direction of the implications). The equivalence concerns the *existence* of a sterile point of $\mathcal{P}'$. For a *given* point x only one direction holds: if x is sterile there is no finite covering, but if x lies in some ET class another point of $\mathcal{P}'$ may still be sterile. An earlier internal draft of [12] said “equivalently” here; this was false and was corrected in review. A weaker statement is available for every explicit point: if $x \in \mathcal{P}'$ lies in no ET class of modulus at most Y , then every finite ET covering of a cofinite subset of $\mathcal{P}$ uses a class of modulus greater than Y . Indeed the classes of modulus $\leq Y$ in the covering miss a neighbourhood of x , which contains infinitely many elements of $\mathcal{P}$.

Remark 2.7 (Heights). The same compactness argument works for any subfamily $\mathcal{K}$ and any function $\kappa \mapsto \text{ht}(\kappa)$. Let $\mathcal{K}_X$ be the triples of height at most X and $U_X = \Sigma \setminus \bigcup_{\kappa \in \mathcal{K}_X} \text{Cl}(\kappa)$ for a compact Σ . The U_X are closed and decreasing. If each is nonempty, the finite intersection property gives a point of $\bigcap_X U_X$, which is $\mathcal{K}$ -sterile. So either some finite subfamily of bounded height covers Σ , or Σ has a $\mathcal{K}$ -sterile point. The set of $\mathcal{K}$ -sterile points of Σ is $\bigcap_X U_X$, a closed set.

3. SQUARE POINTS AND THE ODD-SQUARE OBSTRUCTION

Proposition 3.1 (PROVED; known). *Let $x \in \widehat{\mathbb{Z}}^\times$ be a square point: $x_q \in (\mathbb{Z}_q^\times)^2$ for every prime q . Then x is sterile.*

Proof. For $x = 1$ this is elementary: if $1 \in \text{Cl}(\kappa)$ then the integer $n = 1$ lies in the class, and Lemma 2.1 gives positive integers with $4 = 1/x + 1/y + 1/z \leq 3$. In general, suppose $x \in \text{Cl}(\kappa)$ with $\text{Cl}(\kappa)$ defined modulo M . For each $q \mid M$ the component x_q is the square of a unit, so by the Chinese remainder theorem there is an integer m prime to M with $m^2 \equiv x \pmod{M}$. The primitive class $m^2 \pmod{M}$ is then contained in the class of κ and is solvable by polynomials (Lemma 2.1). This contradicts the theorem of Mordell and Schinzel that a primitive class whose residue is a perfect square is not solvable by polynomials [5, 15], [2, p. 8]. $\square$

The last step is the classical obstruction; ET also derive it from their vanishing result at odd squares [2, Prop. 1.6], which goes back to Schinzel and Yamamoto [17]. The input is one quadratic character computation; Bright and Loughran [1] identify it with the Brauer–Manin obstruction of the Erdős–Straus surfaces.

Call a prime p *Mordell-hard* if $p \pmod{840}$ is a square, $p \pmod{840} \in \{1, 121, 169, 289, 361, 529\}$. Every primitive class modulo 840 that is not a square is solvable by polynomials (Mordell; see [2, p. 8]). The set $\Sigma_{\text{hard}} \subset \widehat{\mathbb{Z}}^\times$ of square classes modulo 840 contains the point 1, so by Propositions 2.5(b) and 3.1 the Mordell-hard primes admit no finite covering by polynomial identities. This is the classical form of the obstruction, and it is the reason why every finite covering statement below restricts the hard primes further by a *non-residue* condition at some additional prime r ; the resulting target sets contain no square point. Whether they contain other sterile points is the question of this note.

3.1. The procedure-level form. The project’s working note [8] proves a version of the odd-square obstruction for procedures. We only state it informally and refer to [8, §§1–2] and [7] for the model of computation. A *witness-producing program* takes a prime p , performs ring operations, floor divisions, eventual-sign comparisons, full factorisations and loops over the lists so produced, and outputs a solution or FAIL; it is *correct* if every solution it outputs is genuine, and *bounded* if the number of steps is bounded

independently of p . For $P \in \mathbb{Z}[X]$ a point $q^* \in \widehat{\mathbb{Z}}$ is *square-mimicking* for P if $P(q_\ell^*)$ is the square of a unit of $\mathbb{Z}_\ell$ for every prime ℓ .

Cited theorem 3.2 (Formal odd-square principle [8, Theorem C, Corollary C1]; CONDITIONAL on Hypothesis H). Let $P = uX + v$ be linear and q^* square-mimicking for P and nondegenerate in the sense of [8, §1.2]. If a correct witness-producing program has a defined and finite formal run at (P, q^*) , then it outputs FAIL at every admissible q (PROVED); under Schinzel’s Hypothesis H [16] for an explicit finite family there are infinitely many admissible q , and $p = P(q) \equiv 1 \pmod{24}$. In particular, under H for one explicit finite family, every correct bounded witness-producing program fails for infinitely many primes $p \equiv 1 \pmod{24}$.

A finite list of ET classes, each tested by a congruence and followed by the identity of Lemma 2.1, is such a bounded program, and its failure at the rational square-mimicking point 1 is Proposition 3.1 for $x = 1$, unconditionally. Cited Theorem 3.2 does not reprove the unconditional statement for general square points; it extends the obstruction from congruence coverings to all bounded procedures, at the price of Hypothesis H. *Novelty*. Theorem C is the procedure-level, H-conditional generalisation of known obstructions (Mordell and Schinzel on square classes, ET Prop. 1.6 and the remark on [2, p. 6] that “a finite set of covering congruence strategies” must fail, Yamamoto, Bright–Loughran); its unconditional instances are known in substance. The project’s novelty audit rates it as a partial novelty only.

Remark 3.3 (ASSESSMENT). The target sets of §§4–?? require x_r to be a non-square for some prime r , so they contain no point that is square-mimicking for $P = X$, and neither Proposition 3.1 nor Cited Theorem 3.2 applies to them. The candidate sterile points below (the sign point $\hat{x}_9$, the point x^* for $r = 13$, the 17-generic line) are therefore not explained by the odd-square obstruction. If they are sterile, the reason is a different, rigidity type of obstruction (§4); this is a heuristic reading, not a theorem.

4. TYPE I COVERINGS AND THE LEAST NON-RESIDUE

4.1. Slices, certificates and the depth $ck_{\min}$. In this section a prime p is *hard* if $p \equiv 1 \pmod{24}$, and n_p denotes the least quadratic non-residue modulo p (a prime ≥ 5). For an odd prime p let

$$\mathcal{B}_p = \{(c, k) : 1 \leq k \leq \lfloor 2p/3 \rfloor, 1 \leq c \leq \lfloor (2p + k)/4k \rfloor, (p, ck) = 1\},$$

put $N_{c,k} = p^2 + 4ck^2$, and let $M_{c,k}(p)$ be the number of divisors D of $N_{c,k}$ with $D \equiv -p \pmod{4ck}$. The Type I solutions of $4/p = 1/x + 1/y + 1/z$ (those with p dividing exactly one denominator) correspond to the pairs $((c, k), D)$ of this kind [6, §§26, 36]; in ET’s coordinates this is the parametrisation $4acd = n + f$, $f \mid 4a^2d + 1$ of family I3 with $(c, d, f)_{\text{ET}} = (k, c, D)$. Write $s = \text{sf}(c)$ for the squarefree part of c and χ_s for the Kronecker character of $\mathbb{Q}(\sqrt{-s})$. If $\chi_s(p) = 1$ then $M_{c,k}(p) = 0$ [6, Thm. 48.1] (genus theory; the

slice is *forced*). For hard p this applies to $s \in \{1, 2, 3, 6\}$, and to every c all of whose prime factors are residues modulo p . Put

$$ck_{\min}(p) = \min\{ck : (c, k) \in \mathcal{B}_p, M_{c,k}(p) > 0\},$$

the least height of a Type I solution in these coordinates. By the forcing just described, $ck_{\min}(p) \geq n_p$ for hard p . For a prime $r \geq 5$ put

$$C(r) = \sup\{ck_{\min}(p) : p \text{ hard}, n_p = r\}, \quad C^*(r) = \limsup_{p \text{ hard}, n_p=r} ck_{\min}(p).$$

A *certificate* is a triple $\kappa = (c, k, F)$ of positive integers with $(F, 4ck) = 1$ and $\text{sf}(c) \notin \{1, 2, 3, 6\}$; its *height* is ck , and its class is the ET class (family I3)

$$\text{Cl}(c, k, F) = \{x \in \widehat{\mathbb{Z}} : x \equiv -F \pmod{4ck}, x^2 \equiv -4ck^2 \pmod{F}\}.$$

If $(c, k) \in \mathcal{B}_p$, then $M_{c,k}(p) \geq 1$ if and only if $p \in \text{Cl}(c, k, F)$ for some F , and the possible F are exactly the divisors counted by $M_{c,k}(p)$ [10, Lemma 0.1]. Let $\Sigma_r \subset \widehat{\mathbb{Z}}^\times$ be the clopen set of x with $x \equiv 1 \pmod{24}$, $x \bmod \ell$ a non-zero square for the primes $5 \leq \ell < r$, and $x \bmod r$ a non-square. By quadratic reciprocity a prime $p > r$ is hard with $n_p = r$ if and only if $p \bmod 24 \prod_{\ell \leq r} \ell$ lies in Σ_r ; by Lemma 2.4 these primes accumulate exactly on Σ_r . A *Type I covering of height $\leq X$* is a finite set of certificates of height $\leq X$ whose classes cover Σ_r ; let $X_r \in \mathbb{N} \cup \{\infty\}$ be the least height of a Type I covering.

The other Type I families I1, I2, I4 also give Type I solutions, but in each of them c or k grows linearly with p [9, §6.1]; only family I3 with fixed (c, k) bounds $ck_{\min}$.

4.2. The case $r = 5$.

Theorem 4.1 (PROVED; [9, Thm. 6.1]). *Every hard prime p with $n_p = 5$ has $ck_{\min}(p) \leq 10$, and $ck_{\min}(p) = 5$ if $p \equiv 2 \pmod{5}$. The bound is attained at $p = 193$, so $C(5) = 10$.*

Proof. Here $p \equiv 1 \pmod{8}$, $p \equiv 1 \pmod{3}$ and $p \equiv 2, 3 \pmod{5}$, so $p \equiv 17$ or $33 \pmod{40}$; all slices below lie in $\mathcal{B}_p$ as $p \geq 73$. If $p \equiv 17 \pmod{40}$, then $-p \equiv 3 \pmod{20}$ and $p^2 + 20 \equiv 0 \pmod{3}$, so $D = 3$ is a target divisor of the slice $(5, 1)$ and $ck_{\min}(p) \leq 5$. If $p \equiv 33 \pmod{40}$, then $-p \equiv 7 \pmod{40}$. As p runs over the units modulo 7, $p^2 \in \{1, 2, 4\}$, while $-20 \equiv 1$, $-40 \equiv 2$, $-80 \equiv 4 \pmod{7}$. So 7 divides one of $p^2 + 20$, $p^2 + 40$, $p^2 + 80$, and $D = 7$ is a target divisor of $(5, 1)$, $(10, 1)$ or $(5, 2)$. Finally $ck_{\min}(193) = 10$ by direct computation, and $ck_{\min}(p) \geq n_p = 5$ always. $\square$

The proof is a Type I covering of Σ_5 of height 10 with $F \in \{3, 7\}$. ES itself is classical for $p \equiv 2, 3 \pmod{5}$; the content is the bounded depth.

4.3. Compactness and Hypothesis H.

Theorem 4.2 ([10, Thm. A]). *Let $r \geq 5$ be prime.*

- (i) (PROVED) *If $X_r < \infty$, then $ck_{\min}(p) \leq X_r$ for every hard p with $n_p = r$ beyond an explicit bound, so $C^*(r) \leq X_r$.*

- (ii) (CONDITIONAL ON H) *If no Type I covering of height $\leq X$ exists, then, under Hypothesis H for an explicit finite family $\mathcal{P}_X$ of polynomials, there are infinitely many hard p with $n_p = r$ and $ck_{\min}(p) > X$.*
- (iii) (PROVED) $X_r = \infty$ *if and only if Σ_r contains a point lying in no class $\text{Cl}(\kappa)$ of a certificate (a Type I-sterile point). Under H, $C^*(r) = X_r$; in particular $C^*(r) = \infty$ if and only if a Type I-sterile point exists.*

Proof sketch. (i) Let $\mathcal{K}$ be a covering of height X_r and $F_{\max}$ its largest F . For a prime $p > \max(2X_r, F_{\max}, r)$ that is hard with $n_p = r$, p is prime to all moduli of $\mathcal{K}$, so its residue is that of a point of Σ_r and lies in some $\text{Cl}(\kappa)$. For $ck \leq X_r < p/2$ one checks $(c, k) \in \mathcal{B}_p$, and $M_{c,k}(p) \geq 1$.

(iii) The first statement is Remark 2.7 with the height ck . The second follows from (i) and (ii).

(ii) By compactness some $x^* \in \Sigma_r$ lies in no class of height $\leq X$. Fix a prime $B \geq \max(X, r, 2m + 2)$, where m is the number of slices of height $\leq X$. At each prime $q \leq B$ at most one value $V = ck^2$ has $x_q^{*2} + 4V = 0$; perturbing x_q^* by a high power of q makes every $v_q(N_{c,k})$ finite and constant on a small class $\mathcal{C} \ni (\text{perturbed point}) \subset \Sigma_r$. Let $f_{c,k}$ be the resulting fixed B -smooth part of $N_{c,k}$. A divisor $d \mid f_{c,k}$ in the target class would give a certificate containing x^* , so none exists. The family $\mathcal{P}_X$ consists of $Qt + a$ (the class $\mathcal{C}$) and the quadratics $((Qt + a)^2 + 4ck^2)/f_{c,k}$; it has no fixed prime divisor. Under H all its members are prime for infinitely many t , and then every divisor of $N_{c,k} = f_{c,k}R_{c,k}$ is d or $dR_{c,k}$ with $d \mid f_{c,k}$; both are excluded from the target class. So $M_{c,k}(p) = 0$ for all slices of height $\leq X$. Full details are in [10, §1]. $\square$

Forcedness is not needed in (ii): forced slices simply contain no certificate class point of Σ_r . A Type I-sterile point need not be sterile in the sense of Definition 2.3; it may lie in a class of another family. Theorem 4.2 concerns the depth $ck_{\min}$, not ES.

Theorem 4.3 (CONDITIONAL ON H; [9, Thm. 2.1]). *Let $g \geq 1$, $D(g) = \sum_{m \leq g} \tau(m)$, and let $r \geq 100D(g)^2g^3$ be prime. Under Hypothesis H for an explicit family of $1 + D(g)$ polynomials, there are infinitely many hard p with $n_p = r$ and $ck_{\min}(p) > gn_p$. Consequently, under H for a finite family depending on ε , $ck_{\min}(p) \geq n_p^{2-\varepsilon}$ for infinitely many hard p .*

The construction is the one of Theorem 4.2(ii) at the point $x_q = 1$ ($q \neq r$), $x_r = b$ with a suitable non-residue b . So no bound $ck_{\min}(p) \leq f(n_p)$ with $f(r) \leq r^{2-\varepsilon}$ for all large r can be proved without refuting H for an explicit family. For fixed r , formal evaluations at H-generic points give explicit lower bounds [9, §6.2]: under H there are infinitely many hard p with $n_p = 7$ and $ck_{\min}(p) = 539$, and with $n_p = 11$ and $ck_{\min}(p) > 3000$. Unconditionally (the same refinement argument plus Dirichlet), every Type I covering of Σ_{11} has height > 3000 (CERTIFIED by finite computation; two independent engines). For $r = 7$ these bounds are superseded by §4.4.

4.4. The sign point.

Lemma 4.4 (PROVED; [10, Lemma 2.1]). *Let $x \in \Sigma_r$ with $x_q \in (\mathbb{Z}_q^\times)^2$ for every prime $q \neq r$. Then a certificate (c, k, F) with $x \in \text{Cl}(c, k, F)$ has $v_r(c)$ odd.*

Proof. $\chi_s(x)$ is determined by $x \bmod 4s$, and equals -1 exactly when $r \mid s$. If $\chi_s(x) = 1$ and $x \in \text{Cl}(c, k, F)$, the open set $\text{Cl}(c, k, F) \cap \{\chi_s = 1\} \cap \Sigma_r$ contains primes p with $(c, k) \in \mathcal{B}_p$ and $M_{c,k}(p) \geq 1$ (Dirichlet), contradicting forcing. $\square$

Definition 4.5. For $w \in \mathbb{Z}_2$ with $w \equiv 9 \pmod{16}$ and a prime $r \equiv 3 \pmod{4}$, the *sign point* $\hat{x}_w = \hat{x}_w^{(r)} \in \widehat{\mathbb{Z}}^\times$ has components w at 2 , -1 at r and 1 at every other prime. For $r = 7$ we write $\hat{x}_9$ for $w = 9$.

The sign point lies in Σ_r because -1 is a non-square modulo r ; for $r \equiv 1 \pmod{4}$ no such point exists. At $\hat{x}_w$ the odd components satisfy $\hat{x}_q^2 = 1$, so $N_{c,k}(\hat{x}) = 1 + 4ck^2$ at every odd prime, and by Lemma 4.4 a certificate at $\hat{x}_w$ is an integer solution of

$$\begin{aligned} F \mid 1 + 4ck^2, \quad F &\equiv -1 \pmod{m'}, \quad F \equiv 1 \pmod{r^{v_r(ck)}}, \\ F &\equiv -w \pmod{2^{v_2(4ck)}}, \quad v_r(c) \text{ odd}, \end{aligned} \quad (1)$$

where m' is the part of ck prime to $2r$. Write $c = 2^\alpha r^a c'$, $k = 2^\gamma r^b k'$ with $c'k'$ prime to $2r$, $t = v_2(4ck) = 2 + \alpha + \gamma$, and $e = (1 + 4ck^2)/F$.

Proposition 4.6 (PROVED; [10, Lemma 3.1, Prop. 4.1]). (a) *If $r \equiv 7 \pmod{8}$, no certificate at $\hat{x}_w^{(r)}$ has $F^2 = 1 + 4ck^2$.*
 (b) *If $r \equiv 3 \pmod{8}$, then $(r(r+1)/4, 2, 2r+1)$ is a certificate at every $\hat{x}_w^{(r)}$, of height $r(r+1)/2$.*

Proof. (b) $v_r(c) = 1$, $1 + 4ck^2 = 1 + 4r(r+1) = (2r+1)^2$, the 2-part of $4ck$ is 8, and $F = 2r+1$ satisfies $F \equiv 1 \pmod{r}$, $F \equiv -1 \pmod{(r+1)/4}$, $-F \equiv 1 \pmod{8}$. (a) If $F^2 = 1 + 4ck^2$, then from (1) $m' \mid F+1$ and $r^{a+b} \mid F-1$; as $\gcd(F-1, F+1) = 2$ this gives $F+1 = 2^i c'k'^2$, $F-1 = 2^j r^{a+2b}$ with $i+j = 2 + \alpha + 2\gamma$, $\min(i, j) = 1$ and $a+2b$ odd. Since $v_2(1+r^s) = 3$ for odd s when $r \equiv 7 \pmod{8}$, the case $j = 1$ forces $\alpha + 2\gamma = 3$ and then $-F \equiv 1 \pmod{16}$, against $w \equiv 9 \pmod{16}$; the cases $i = 1$ lead to $-F \not\equiv 1 \pmod{8}$ or to a parity contradiction. See [10, Lemma 3.1]. $\square$

So sign points survive the square family exactly for $r \equiv 7 \pmod{8}$: $r = 7, 23, 31, 47, \dots$. At the other natural candidate, the *residue-one point* ($x_q = 1$ for $q \neq 7$), every non-square x_7 is already covered by $(7, 3, 11)$, $(7, 3, 23)$ or $(14, 2, 15)$ [10, Prop. 2.3].

Lemma 4.7 (PROVED; [11, Lemmas 1.1–1.2]). *Let (c, k, F) be a certificate at $\hat{x}_w$ and $f \in \{F, e\}$. Then $m' \mid f+1$, $r^{a+b} \mid f-1$, $f \equiv -w \pmod{2^t}$ if $f = F$ and $wf \equiv -1 \pmod{2^t}$ if $f = e$, and $4ck^2 \equiv -1 \pmod{f}$. Conversely such data with a odd and $a, c', k' \geq 1$ give a certificate. If w is a positive integer,*

each f belongs to finitely many certificates, all explicitly computable from the factorisation of $f + 1$. If $ck \leq X$ and $r \mid c$, then $\min(F, e) < 2X/\sqrt{r} + 1$.

The finiteness needs w with finite 2-adic depth $v_2(f + w)$, $v_2(wf + 1)$: at $w = -71$ the triples $(14 \cdot 2^{35j}, 8192, 71)$ are certificates for all $j \geq 0$. For $w = 9$ the lemma reduces the search to a sieve over f , at all heights simultaneously.

Computation 4.8 (CERTIFIED; [10, Comp. 3.2], [11, Comp. 2.1, 2.3]). (a)

No certificate at $\hat{x}_9$ ($r = 7$) has $\min(F, e) < 10^{12}$. Two independent engines (`typei3_fsearch.c` and the reviewer's `review_typei3_fs.c`) cover $f < 10^{11}$; the range $[10^{11}, 10^{12})$ rests on one engine. An earlier height-graded search (`typei2_signcheck.c`, $7.6 \cdot 10^9$ slices with $ck \leq 3 \cdot 10^9$) was independently re-run to its full range, and cross-checked by a factoring engine and a complete parametric enumeration on smaller ranges.

(b) For $r = 23, 31, 47$ and $w = 9$ no certificate has $\min(F, e) < 10^{11}$ (two engines for $r = 23$, one for $r = 31, 47$).

On ten test points with known certificates the f -graded and the height-graded engines produce identical certificate sets.

Corollary 4.9. (CERTIFIED) *Every Type I covering of Σ_7 has height $> 1.32 \cdot 10^{12}$; for $r = 23, 31, 47$ every Type I covering of Σ_r has height $> 2.39 \cdot 10^{11}$, $> 2.78 \cdot 10^{11}$, $> 3.42 \cdot 10^{11}$ respectively. (CONDITIONAL ON H) There are infinitely many hard p with $n_p = 7$ and $ck_{\min}(p) > 1.32 \cdot 10^{12}$; so $C(7) > 1.32 \cdot 10^{12}$, and similarly for $r = 23, 31, 47$.*

Proof. By Lemma 4.7, a certificate of height $\leq X$ at $\hat{x}_9$ has $\min(F, e) < 2X/\sqrt{7} + 1$; with Computation 4.8 there is none with $X \leq 1.32 \cdot 10^{12}$. A covering of Σ_7 must contain $\hat{x}_9$. The conditional part is Theorem 4.2(ii) with the point $x^* = \hat{x}_9$. $\square$

The previous bounds were $C(7) \geq 539$ [9] and $C(7) > 3 \cdot 10^9$ [10]; the census maximum of $ck_{\min}$ over hard $p < 10^7$ with $n_p = 7$ is 76 (EVIDENCE).

4.5. Low 2-adic levels. The certificates at the sign point can also be attacked by descent.

Lemma 4.10 (Vieta descent; PROVED; [11, Lemma 5.1]). *Let $B, K, \delta, F, e \geq 1$ be integers with $Fe = 1 + BK^2$ and $e - F = BK\delta$. Then $F \equiv e \equiv 1 \pmod{B\delta}$.*

Proof. Induction on K . If $F = 1$ then $K = \delta$ and $e = 1 + B\delta^2$. Otherwise put $\rho = K - \delta F$. From $F^2 + BK\delta F = 1 + BK^2$ we get $F^2 - 1 = BK\rho$, so $1 \leq \rho < K$. Put $F' = F - B\rho\delta$; then $F'F = 1 + B\rho^2 > 0$ and $F - F' = B\rho\delta$, and induction applies to (F', F) . $\square$

Proposition 4.11 (PROVED; [11, Cor. 5.2, Props. 5.3–5.5]). *Let $w \equiv 9 \pmod{16}$ and $r = 7$. Every certificate at $\hat{x}_w$ has $t \geq 5$ and $\alpha + 2\gamma \geq 7$.*

Sketch. By Lemma 4.7 and $w \equiv w^{-1} \pmod{16}$, $e \equiv F$ modulo $2^{\min(t, 4)}$ times the odd part of ck , and $F, e \not\equiv 1 \pmod{4}$; $F \neq e$ by Proposition 4.6(a). If

$t \leq 4$, Lemma 4.10 with $B = 4c$ gives $F \equiv 1 \pmod{4}$; if $\alpha + 2\gamma \leq 4$ the same holds after the integral rescaling $c \mapsto c/4^{t-4}$, $k \mapsto 2^{t-4}k$. At levels $\alpha + 2\gamma \in \{5, 6\}$ the descent stays integral and forces $c = 2^\alpha 7^a$. At level 5 the chain of the descent, read modulo 16, only reaches divisors $\equiv \pm 1 \pmod{16}$, while certificates need $\equiv 7$; this deduction is due to the internal reviewer. At level 6 the chain is a Pell orbit; a 2-adic index condition forces its index to be an odd multiple of 3, whence $D + 3$ divides the relevant Lucas number ($D = 7^a \delta^2$), and a prime $q \mid (D + 3)/2$ gives $F \equiv 1$ and $F \equiv -1 \pmod{q}$ simultaneously. Details and machine checks are in [11, §5]. $\square$

At level $\alpha + 2\gamma = 7$ the descent leaves $\mathbb{Z}$, and 34 near misses (certificates except for the 2-adic condition) with $-F \equiv 1 \pmod{8}$ and $ck \leq 10^8$ exist there; this level is open.

4.6. Scope of a sterility proof.

Proposition 4.12 (PROVED; [11, Prop. 3.1]). *The set of Type I-sterile points of Σ_7 is closed and has empty interior.*

Proof. Closedness is Remark 2.7. Let $U = \{y \equiv x \pmod{Q}\} \subset \Sigma_7$ be a basic neighbourhood, $Q = 2^j 7^i \prod_{q \in P} q^{e_q}$. Pick a prime $p_0 \notin P \cup \{2, 7\}$ and, by Dirichlet, a prime $F \notin P \cup \{2, 7, p_0\}$ with $F \equiv 3 \pmod{4}$, $F \equiv -x_7 \pmod{7}$ and $(-7p_0/F) = 1$; these are compatible because $(-7/F) = (F/7) = (-x_7/7) = 1$ for non-square x_7 . Then $\text{Cl}(7p_0, 1, F)$ meets U by the Chinese remainder theorem. $\square$

So no sterility proof can consist of exhibiting one ambient congruence cylinder: a sterile point, if it exists, is a limit of covered points. This does not exclude arguments within the thin sign fibre $\Phi = \{\hat{x}_w : w \in 9 + 16\mathbb{Z}_2\}$, which itself has empty interior.

Remark 4.13 (Measure route; PROVED reduction and EVIDENCE). Theorem 4.2(iii) needs *some* Type I-sterile point, not $\hat{x}_9$. By Lemma 4.7, the certificates having a given $f \equiv 7 \pmod{16}$ remove from Φ at most two 2-adic balls of radius $2^{-t_{\min}(f)}$, with $t_{\min}(f) \geq 5$ by Proposition 4.11; other f remove nothing. Let U_Y be the part of Φ surviving all $f < Y$. If

$$\mu(U_Y) > \sum_{f \geq Y} 2^{5-t_{\min}(f)}$$

(Haar measure on w , normalised to 1), then Φ contains a Type I-sterile point, and $C^*(7) = \infty$ under H. Numerically $\mu(U_{10^{10}}) \geq 0.6086$, the bound including the error of truncating the 2-adic depth at 40; and $t_{\min}(f)$ grows like $\frac{1}{2} \log_2 f$ in the data (EVIDENCE). An explicit summable bound for the number of f with given $t_{\min}$ would therefore close the question; a bound $o(2^T)$ alone does not suffice. No such bound is known.

Conjecture 4.14 ([10, Conj. 3.4]). *The sign point $\hat{x}_9$ is Type I-sterile. Then, under H, $C(7) = C^*(7) = \infty$.*

The evidence is the search range of Computation 4.8, the empty levels of Proposition 4.11, and the decay of the near-miss mass (EVIDENCE). Among integral points $(x_2, x_7) = (w, z)$ with $w \equiv 9 \pmod{16}$, $|w| \leq 200$, $|z| \leq 60$ (all other components 1), the 26 that survive the complete enumeration of certificates at $\{2, 7\}$ -generic points up to the cutoff of [10, §2.2] (parameter $\Lambda \leq 3 \cdot 10^8$ there) all have $z = -1$; this is how the sign point was found. For $r \equiv 1 \pmod{4}$ no sign point exists, and $r = 5$ is the one value with a known finite covering.

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PROVENANCE-PLACEHOLDER

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